

5.1.4 Hormonal communication

The ways in which specific hormones bring about their effects are used to exemplify endocrine communication and control. Treatment of diabetes is

used as an example of the use of medical technology in overcoming defects in hormonal control systems.

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) endocrine communication by hormones	To include secretion of hormones into the blood, transport by the blood, and detection by target cells or tissues.
(b) the structure and functions of the adrenal glands	Adrenal glands as an example of endocrine glands, to include the hormones secreted by the cortex and medulla and their functions.
(c) (i) the histology of the pancreas (ii) the examination and drawing of stained sections of the pancreas to show the histology of the endocrine tissues	To include the endocrine tissues. PAG1 HSW4
(d) how blood glucose concentration is regulated	To include the action of insulin and glucagon as an example of negative feedback, and the role of the liver AND the control of insulin secretion, with reference to potassium channels and calcium channels in the beta cells of the pancreas. HSW12
(e) the differences between Type 1 and Type 2 diabetes mellitus	To include the causes of Type 1 and Type 2 diabetes and the treatments used for each. HSW12
(f) the potential treatments for diabetes mellitus.	To include the use of insulin produced by genetically modified bacteria and the potential use of stem cells to treat diabetes mellitus. HSW12